

The motivating cusp symbol in arXiv:2601.11934 is already locally Wiener

Candidate partial-result packet

11 August 2026

Status. Candidate substantial partial result; likely valid; human review recommended. It settles the explicit symbol displayed in Remark 1.5 of the source, but it does not remove the Wiener hypothesis for every C^m symbol.

1 Source question and claimed obstruction

Chen and Hong study nonlinear superposition operators on quantum Besov spaces in their 2026 preprint [1]. Their Theorem 1.4 assumes, locally, $F \in W_0(\mathbb{R}) \cap W_{\lceil s \rceil}(\mathbb{R})$. Remark 1.5 states that when $\lceil s \rceil = 2k + 1$ and

$$F(x) = |x|x^{2k+1},$$

this symbol is not locally in $W_0 \cap W_{2k+1}$, and presents it as the motivating example for the regularity gap. The source passage is reproduced below (PDF page 6).

result only requires $F \in C^r(\mathbb{R})$, $r > s$, whereas our result requires $F \in C^{\lceil s \rceil}(\mathbb{R})$. Second, in the noncommutative setting, the conditions imposed on F for $1 < p < \infty$ and for $p = 1, \infty$ differ substantially. This distinction originates from the boundedness properties of operator integrals on noncommutative L_p spaces, which will be discussed in Section 4.

When $s \leq \frac{d}{p}$, the lack of embedding $B_{p,q}^s(\mathbb{R}_\theta^d) \hookrightarrow L_\infty(\mathbb{R}_\theta^d)$ forces us to consider the boundedness estimate of T_F on the multiplication algebra $B_{p,q}^s(\mathbb{R}_\theta^d) \cap L_\infty(\mathbb{R}_\theta^d)$. With more regularity assumption on F , we obtain the following local boundedness estimate of T_F on $B_{p,q}^s(\mathbb{R}_\theta^d) \cap L_\infty(\mathbb{R}_\theta^d)$, $s > 0$, which follows from Theorem 1.3 when $s > \frac{d}{p}$. Let $W_n(\mathbb{R})$ denote the Wiener space, where the definition can also be seen in Section 4.

Theorem 1.4. *Let $1 \leq p, q \leq \infty$, $s > 0$ and $u \in B_{p,q}^s(\mathbb{R}_\theta^d) \cap L_\infty(\mathbb{R}_\theta^d)$ self-adjoint, if $F \in_{\text{loc}} W_0(\mathbb{R}) \cap W_{\lceil s \rceil}(\mathbb{R})$ and $F(0) = 0$, then we have*

$$(1.11) \quad \|F(u)\|_{B_{p,q}^s} \lesssim_{s,p,\|u\|_\infty} \|F\|_{W_0 \cap W_{\lceil s \rceil}}^{\|u\|_\infty, \text{loc}} \|u\|_{B_{p,q}^s}.$$

Remark 1.5. Theorem 1.4 requires more regularity assumption on F than Theorem 1.3. For example, when $\lceil s \rceil = 2k + 1$ for some $k \in \mathbb{N}^+$ and $F = |x|x^{2k+1}$, we then have $F \in C^{2k+1}(\mathbb{R})$, whereas $F \notin_{\text{loc}} W_0(\mathbb{R}) \cap W_{2k+1}(\mathbb{R})$ due to the singularity of the Hilbert transform on $L_1(\mathbb{R})$. Unfortunately, we do not know how to remove this gap.

On PDF page 23, the source defines

$$W_n(\mathbb{R}) = \{G \in C^n(\mathbb{R}) : G \in L_\infty(\mathbb{R}), F(G^{(n)}) \in L_1(\mathbb{R})\}.$$

Membership is local when every smooth compact localization has this property. Under that definition, the displayed nonmembership assertion is false.

2 Idea of the proof

The key observation is one-dimensional and elementary. A compactly supported H^1 function has integrable Fourier transform: insert the weight $(1 + \xi^2)^{-1/2}$ and use Cauchy–Schwarz and Plancherel. For the cusp symbol, the allegedly troublesome top derivative is just a constant multiple of $|x|$. After a compact cutoff it is Lipschitz, hence in H^1 . The same is true of the localized original function, so both W_0 and W_m conditions hold. This also yields a useful sufficient class substantially broader than the single example.

3 A local Wiener criterion

We use any standard unitary or nonunitary normalization of the Fourier transform; only harmless constants change.

Lemma 1. *If $h \in H^1(\mathbb{R})$, then $\widehat{h} \in L_1(\mathbb{R})$ and*

$$\|\widehat{h}\|_1 \leq C \|h\|_{H^1}.$$

Proof. By Cauchy–Schwarz and Plancherel,

$$\begin{aligned} \int_{\mathbb{R}} |\widehat{h}(\xi)| d\xi &\leq \left(\int_{\mathbb{R}} \frac{d\xi}{1 + \xi^2} \right)^{1/2} \left(\int_{\mathbb{R}} (1 + \xi^2) |\widehat{h}(\xi)|^2 d\xi \right)^{1/2} \\ &\leq C \|h\|_{H^1}. \end{aligned}$$

□

Theorem 1. *Let $m \geq 1$. Suppose that $F \in C^m(\mathbb{R})$ and $F^{(m)} \in H_{\text{loc}}^1(\mathbb{R})$. Then*

$$F \in_{\text{loc}} W_0(\mathbb{R}) \cap W_m(\mathbb{R}).$$

In particular, the conclusion holds whenever $F^{(m)}$ is locally Lipschitz.

Proof. Let $\chi \in C_c^\infty(\mathbb{R})$ and put $G = \chi F$. The function G is bounded, compactly supported, and belongs to C^m . Since $m \geq 1$, it is also a compactly supported C^1 function, hence $G \in H^1(\mathbb{R})$. Lemma 1 gives $\widehat{G} \in L_1$, which is the W_0 condition.

Leibniz’s formula gives

$$G^{(m)} = \sum_{j=0}^m \binom{m}{j} \chi^{(m-j)} F^{(j)}.$$

The term with $j = m$ lies in H^1 : multiplication by a smooth compactly supported function preserves local H^1 . If $j < m$, then $F^{(j)}$ is locally C^1 , so $\chi^{(m-j)} F^{(j)}$ is compactly supported C^1 and therefore is in H^1 . Thus $G^{(m)} \in H^1(\mathbb{R})$. A second application of Lemma 1 yields $\widehat{G^{(m)}} \in L_1$, which is the W_m condition. Since χ was arbitrary, the membership is local. □

4 The cusp in Remark 1.5

Corollary 1. *Let $m = 2k + 1$ be odd and set $F(x) = |x|x^m$. Then*

$$F \in_{\text{loc}} W_0(\mathbb{R}) \cap W_m(\mathbb{R}).$$

Consequently, whenever $\lceil s \rceil = m$, this symbol already satisfies the symbol hypothesis of Theorem 1.4 in [1]; hence that theorem gives its stated quantum Besov estimate for all of its advertised p, q, s and bounded self-adjoint inputs.

Proof. On the positive and negative half-lines,

$$F(x) = \begin{cases} x^{m+1}, & x > 0, \\ -x^{m+1}, & x < 0. \end{cases}$$

Differentiating m times gives, on both half-lines,

$$F^{(m)}(x) = (m+1)!|x|.$$

Both sides are continuous at zero, so this is the classical m -th derivative on all of $\mathbb{R}$. Its weak derivative is $(m+1)!\operatorname{sgn}(x)$ almost everywhere, which belongs to L_2 on every compact interval. Therefore $F^{(m)} \in H_{\text{loc}}^1$, and Theorem 1 applies. Finally $F(0) = 0$, so the remaining symbol hypothesis of the source theorem is also satisfied. $\square$

5 Verification and limitations

The proof was checked directly against the current arXiv v2 definition of W_n . It uses only Leibniz’s formula, Cauchy–Schwarz, and Plancherel. The source crop contains the whole of Theorem 1.4 and Remark 1.5, with no omitted continuation. No computational experiment is being presented as evidence.

The result does not show that every C^m function is locally in W_m . A merely continuous compactly supported function need not have integrable Fourier transform, so the hypothesis $F^{(m)} \in H_{\text{loc}}^1$ is a real additional condition. Accordingly, the general problem of replacing local Wiener regularity by the minimal interior- p hypotheses remains open here.

A bounded novelty search on 11 August 2026 queried the exact title and arXiv identifier, the exact cusp symbol, “remove this gap”, “quantum Besov”, and Wiener-space phrases. It found the current source but no later arXiv correction or resolution of this observation. Novelty confidence is therefore moderate, not definitive.

Human-review recommendation. Check the cutoff convention for $\in_{\text{loc}}$ and the source’s Fourier normalization. Neither affects the argument. If confirmed, Remark 1.5’s specific example should be corrected, while its broader regularity question should be retained.

References

- [1] D. Chen and G. Hong, *The nonlinear estimates on quantum Besov spaces*, arXiv:2601.11934v2, 2026.